

Question and experiment

Fine-tune pretrained Qwen3.5 base language models at 0.8B, 2B, and 4B. Compare direct contacts-v1 documents against plain-language instructions, ordinary amino-acid sequences, and one-based contact pairs. Six trials, approximately 1B tokens each; at most 48 H100 GPUs.

Data and comparison

Use exp232's existing CoreWeave copy of exp225-decontaminated AFDB documents. The two formats contain the same sequence and contact information. Keep Qwen's original tokenizer and vocabulary. No chat template. Exclude a protein from both formats if either complete document exceeds 16K. Hold out sequence clusters, with no use of eval-test for model development.

Current evidence and limits

The complete corpus contains 920,611 training and 9,235 validation documents. 5.11% of source examples exceeded 16K; both formats exclude the same rows. The training pool contains 4.493B contacts-v1 tokens and 1.290B prompted tokens. Equal token budgets imply different protein exposure. The 0.8B training, checkpoint, and optimizer/data/RNG recovery checks passed. Both 0.8B production formats have been launched; the 2B prompted smoke is training. No contact accuracy or pretrained-transfer advantage has yet been established.

Validation and interpretation

Teacher-forced contact likelihood uses 64 held-out AFDB documents. Greedy generation diagnostics retain malformed, empty, and capped outputs. They are not the established FoldBench rollout-plus-resample benchmark. Per-input timings are saved at evaluation time, including worker metadata. The first raw-format run initially sustained roughly 39–53K tokens/second over eight H100s. These are engineering observations, not accuracy results.

validation_nll.png

Production validation on 64 held-out AFDB documents. Lower is better within a format; token NLL across formats is not contact accuracy. In-progress observations, without error bars.

